

2 **Goodness-of-fit for multi-distribution neutrino** 3 **cross-section measurements with shared events**

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10 **ABSTRACT:** When multiple differential cross-section measurements are extracted from a common
11 event sample, the event-sharing structure between distributions imposes exact linear constraints
12 among the bin counts that reduce the effective dimensionality of the measurement below the total bin
13 count. When the statistical covariance matrix accounts for these inter-distribution correlations, the
14 constraints manifest as a rank deficiency. Systematic uncertainties may formally restore invertibility,
15 but the resulting near-singular directions yield unphysically inflated χ^2 values in global goodness-
16 of-fit tests. We present the range-projected χ^2 , a test statistic that restricts the goodness-of-fit test to
17 the subspace carrying independent statistical information, yielding a χ^2 with $N_{\text{bins}} - N_{\text{null}}$ degrees of
18 freedom, where N_{null} is the number of independent constraints. We show that this rank deficiency
19 is a predictable consequence of the event-sharing structure, and that N_{null} decomposes into a
20 structural contribution determined a priori from the binning geometry and a kinematic contribution
21 that depends on the phase-space occupancy. The method is validated with an analytical toy model
22 and a simulated neutrino–argon cross-section measurement including unfolding and multi-source
23 systematic uncertainties.

24 **KEYWORDS:** Analysis and statistical methods

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49 1 Introduction

50 Modern neutrino cross-section literature is experiencing an increase of published measurements
51 reporting multiple differential distributions extracted from the same event sample. MicroBooNE
52 has reported simultaneous measurements of cross sections for final states with and without protons
53 [1, 2], multi-differential QE-like cross section measurements in kinematic imbalance variables
54 [3, 4], and triple-differential inclusive results [5]. Other relevant examples include the multiple
55 final-state kinematic projections from T2K [6, 7], and double-differential inclusive results from
56 NOvA [8–10]. In each of these analyses, the same selected events populate bins in multiple

reported distributions, leading to strong inter-measurement statistical correlations. Separate but related efforts have produced simultaneous measurements across different nuclear targets [11–13] and combined neutrino-antineutrino results [14], where correlations arise from shared systematic uncertainties rather than shared events.

When multiple distributions are extracted from a common event sample, the event-sharing structure imposes exact linear constraints among the bin counts. If every selected event populates one bin in each distribution, then the total number of events in all of them is exactly identical. More generally, the pattern of shared bin occupancy across distributions determines a set of constraints that reduce the effective dimensionality of the measurement below the total bin count. These constraints are a property of the measurement itself, independent of any particular statistical treatment.

The blockwise unfolding framework of Ref. [15] provides a rigorous method for computing the full cross-distribution covariance matrices in such measurements. In this approach, all bins corresponding to the same measured observable in the multi-distribution analysis are organized in blocks, each corresponding to a single differential cross-section distribution. Each bin block is unfolded independently using any standard method (e.g. iterative D’Agostini unfolding [16] or Wiener-SVD [17]), while the full covariance matrix, encoding the correlations between bins from different distributions, is constructed and propagated through the cross section extraction procedure. The statistical covariance for bins in different blocks is determined by the number of events they share. The constraints described above can manifest as a rank deficiency in the statistical covariance matrix, as certain directions in bin space carry identically zero variance. The number of these degenerate directions is exactly predictable from the binning geometry.

A natural use of these combined measurements is computing global goodness-of-fit metrics, such as a χ^2 statistic, that test a model against all distributions simultaneously. Compared to evaluating each distribution independently, a global test that accounts for all inter-block correlations uses the full inter-block statistical information of the data. Such global metrics are essential for interaction model tuning efforts [18–20], where measurements of multiple kinematic distributions provide complementary sensitivity to different aspects of the underlying physics. However, a global χ^2 that does not account for the constrained directions yields a test with the wrong effective number of degrees of freedom. When the statistical covariance matrix is constructed under the Poisson assumption, the constraints appear as exact null directions that prevent direct matrix inversion. Systematic uncertainties may formally restore invertibility, but the resulting near-singular directions produce unphysically inflated χ^2 values. In either case, the test is unreliable unless the constrained directions are properly identified and excluded.

We present a method to correctly compute a global goodness-of-fit test statistic for multi-distribution measurements with shared events. The approach identifies the constrained directions from the bin-combination structure of the measurement and projects the statistic onto the subspace where independent statistical information is available; we refer to the resulting test statistic as the range-projected χ^2 . Our contribution is twofold: we show that the rank deficiency of the statistical covariance is a predictable consequence of the shared-event structure, rather than an incidental numerical degeneracy, and we provide a prescription to compute the number of constrained directions a priori from the binning geometry. The statistical foundation of the projected statistic is the standard treatment of quadratic forms with singular covariance matrices, a connection we make explicit while contrasting our approach with related methods.

100 The paper is organized as follows. Section 2 describes the constraints imposed by the shared-
 101 event structure; Sec. 3 constructs the range-projected test statistic and relates it to existing ap-
 102 proaches; Sec. 4 and Sec. 5 validate the method on a toy model and a simulated measurement; and
 103 Sec. 6 discusses the scope and limitations of the method.

104 2 Constraints from shared events

105 Consider a neutrino cross-section analysis simultaneously measuring B different flux-integrated
 106 differential cross sections, each with N_b true bins. If the measurement is extracted from a common
 107 event sample then each selected event will populate exactly one bin in each distribution it belongs
 108 to. Recent MicroBooNE analyses [21] have adopted the blockwise unfolding approach presented
 109 in Ref. [15] to simultaneous cross-section measurements.

110 This method provides a systematic framework for reporting the correlations between the dif-
 111 ferent extracted differential cross sections. It provides a general prescription to construct statistical
 112 covariance matrices that respect the correlations between bins in different distributions or blocks.
 113 The full inter-block covariance matrix constructed in reconstructed space is propagated through
 114 unfolding using a block-diagonal unfolding matrix, obtained as a direct sum of the individual un-
 115 folding matrices computed for each block. The result of the combined measurement is a vector of
 116 $N_{\text{bins}} = \sum_{b=1}^B N_b$ cross-section values with an $N_{\text{bins}} \times N_{\text{bins}}$ covariance matrix. The total covariance
 117 decomposes as $C_{\text{total}} = C_{\text{stat}} + C_{\text{syst}}$, where the statistical covariance component, computed from the
 118 shared-event structure, encodes the constraints described in the introduction.

119 2.1 Statistical covariance from shared events

120 The core idea behind the treatment of the statistical correlations in the blockwise unfolding approach
 121 is the partition of any arbitrary pair of overlapping kinematic bins X and Y into a set of three non-
 122 overlapping bins u , v , and w . The events in the intersection $X \cap Y$ are assigned to bin v , while bin
 123 u contains the events in the relative complement of Y in X (i.e. $X \setminus Y$), and bin w corresponds to
 124 the relative complement of X in Y (i.e. $Y \setminus X$). Denoting the number of predicted events in each
 125 bin as n_i , and assuming these follow independent Poisson distributions, the covariance between the
 126 counts in bins X and Y is given by [15]

$$\begin{aligned} \text{Cov}(n_X, n_Y) &= \text{Cov}(n_u + n_v, n_v + n_w) \\ &= \text{Cov}(n_u, n_v) + \text{Cov}(n_u, n_w) + \text{Cov}(n_v, n_v) + \text{Cov}(n_v, n_w) \\ &= \text{Var}(n_v), \end{aligned} \tag{2.1}$$

127 as by definition bins u , v , and w have no events in common. This implies that the covariance
 128 between any two bins can be estimated from the variance of the events they share.

129 The variance for a set of Monte Carlo (MC) events can be estimated from the sum of their
 130 squared weights. This gives us a general prescription to calculate the elements of the statistical
 131 covariance matrix as

$$(C_{\text{stat}})_{ij} = \sum_{e \in (i \cap j)} w_e^2, \tag{2.2}$$

where w_e is the weight associated to event e belonging to the intersection of bins i and j . In the case of measured data the weights are 1, and the covariance element simplifies to the number of events shared between the bins n_{ij} .

After propagation through the block-diagonal unfolding, the statistical covariance matrix in unfolded space inherits the same rank structure, as the unfolding matrices for each individual block are invertible. This rank structure has a direct interpretation in terms of the number of distinct event-sharing patterns across distributions, which we discuss next.

2.2 Combination matrix and null space

The previous discussion shows that the statistical covariance between any two bins is determined solely by the events they share. We now formalize this observation to derive an exact formula for the rank of C_{stat} , and therefore for the number of directions in bin space that carry zero statistical variance.

For each event e , we can define the indicator vector $\phi_e := (\phi_{e,i})_{i=1}^{N_{\text{bins}}}$ such that $\phi_{e,i} = 1$ if $i \in \text{bins}(e)$ and $\phi_{e,i} = 0$ if $i \notin \text{bins}(e)$, where $\text{bins}(e)$ is the set of bins event e populates across all distributions. The indicator vector acts as an identifier of each event in the combined measurement. For example, in a two-block measurement with 3 bins each, an event populating bin 2 of block 1 and bin 1 of block 2 has indicator vector $\phi_e = (0, 1, 0, 1, 0, 0)$. The statistical covariance matrix can then be re-written in terms of indicator vectors as

$$C_{\text{stat}} = \sum_e w_e^2 \phi_e \phi_e^T. \quad (2.3)$$

This outer-product decomposition makes the rank structure of C_{stat} clear. The rank of the statistical covariance matrix equals the dimension of $\text{span}\{\phi_e\}$, which depends only on the distinct indicator vectors present, not on their weights. Since all weights are $w_e > 0$ no indicator vector is removed from the sum, and the column space of C_{stat} is exactly $\text{span}\{\phi_e\}$.

We define the bin-combination matrix Ω as the matrix whose rows are the unique indicator vectors among $\{\phi_e\}$. Each row of Ω represents a distinct pattern of bin occupancy across the blocks. By construction the row space of Ω equals $\text{span}\{\phi_e\}$, and therefore

$$\text{rank}(C_{\text{stat}}) = \text{rank}(\Omega). \quad (2.4)$$

The number of degenerate directions is then

$$N_{\text{null}} = N_{\text{bins}} - \text{rank}(\Omega). \quad (2.5)$$

Physically, each of the N_{null} directions corresponds to a linear combination of bins along which the measurement has identically zero variance. Any component of the residual vector along such a direction is not perceived by the statistical covariance; it is a direction that the measurement has no statistical power to constrain. Importantly, both Ω and N_{null} are directly computable from the MC event-to-bin mapping alone, before constructing any covariance matrix, confirming that the constraints are a property of the measurement geometry rather than the statistical treatment.

A vector $\mathbf{x}$ is in the null space of C_{stat} if and only if, for every event e

$$\sum_{i \in \text{bins}(e)} x_i = 0, \quad (2.6)$$

or equivalently $\Omega \mathbf{x} = 0$. Each unique indicator vector acts as a linear constraint on the null-space vectors, requiring that the components of $\mathbf{x}$ sum to zero over the bins populated by the different events. The null space directions can be divided into two physically distinct categories: structural constraints that follow from the block topology alone, and kinematic constraints that depend on which regions of phase space are populated.

2.3 Determination of the structural null space

The combination matrix Ω presented previously is constructed from MC events. One determines which events populate the different bins and collects the unique indicator vectors. However, the indicator vectors that can appear are constrained by two distinct mechanisms. The first is purely structural, the geometry of the binning scheme forbids certain bin combinations. The second is kinematic, correlations between the different variables render some combinations unreachable even though they are geometrically allowed. These constraints are imposed by the underlying physics effects, mainly the differential cross section and the detector acceptance. In the following, we show that the structural constraints can be determined a priori, without generating MC events, from the bin edges used.

The variables used in a measurement are drawn from a set $\{v_1, \dots, v_D\}$. Each block b uses a subset of these variables, $\text{vars}(b) \subseteq \{v_1, \dots, v_D\}$. Restricting the discussion to combinations of single- and double-differential measurements, we have $|\text{vars}(b)| \in \{1, 2\}$ for every b . For each variable v used in a given block b , the binning scheme defines a set of bin edges $E_v^{(b)}$ that partition the range of v into intervals. In the case of a two-dimensional block, the edges in the binning variable may be slice-dependent, but nevertheless these can be collected into a single set of edges per variable across slices. For each variable $v \in \bigcup_b \text{vars}(b)$, we can define the refined edges

$$\mathcal{E}_v = \bigcup_{b: v \in \text{vars}(b)} E_v^{(b)}, \quad (2.7)$$

with $M_v = |\mathcal{E}_v| - 1$ denoting the number of refined bins for variable v . Since $\mathcal{E}_v$ contains all the bin edges from every block that uses that variable, each refined bin is contained entirely within one original bin of every such block.

The common refinement of all blocks is the Cartesian product of the refined bins across all variables, denoted by $\mathcal{R}$. Each cell $c \in \mathcal{R}$ specifies a refined bin index for every variable in the measurement. For every block b , there exists a unique mapping $\beta_b(c)$ between each cell c and an original bin in said block. The elements of the indicator vector $\phi_c := (\phi_{c,i})_{i=1}^{N_{\text{bins}}}$ associated to cell c can be written as

$$\phi_{c,i} = \begin{cases} 1 & \text{if } i = \beta_b(c) \text{ for any block } b, \\ 0 & \text{otherwise.} \end{cases} \quad (2.8)$$

If block b does not use variable v then β_b is independent of the refined bin index in v . Therefore, multiple cells may produce identical indicator vectors.

The structural combination matrix Ω_{struct} is defined as the matrix whose rows are the unique indicator vectors from the cells of the common refinement $\{\phi_c : c \in \mathcal{R}\}$. By construction, every indicator vector that appears in the Ω matrix obtained from the MC events is also present in Ω_{struct} , therefore

$$\text{rank}(\Omega) \leq \text{rank}(\Omega_{\text{struct}}) \leq N_{\text{bins}}. \quad (2.9)$$

201 This hierarchy motivates a decomposition of the total number of degenerate directions

$$\begin{aligned}
N_{\text{null}} &= N_{\text{bins}} - \text{rank}(\Omega_{\text{struct}}) + \text{rank}(\Omega_{\text{struct}}) - \text{rank}(\Omega) \\
&= N_{\text{null}}^{\text{struct}} + N_{\text{null}}^{\text{kinem}}.
\end{aligned}
\tag{2.10}$$

202 The first term, $N_{\text{null}}^{\text{struct}}$, counts null directions that arise purely from the binning geometry and is
203 computable a priori from the bin edges alone. The second term, $N_{\text{null}}^{\text{kinem}}$, accounts for additional null
204 directions introduced by phase-space restrictions that exclude geometrically allowed combinations
205 and can only be determined from Monte Carlo. A description of the algorithm used to compute
206 Ω_{struct} is presented in Appendix A.

207 The presence of kinematic null directions indicates that certain regions of the combined bin
208 space are empty, which may signal that the binning extends beyond the kinematic reach of the
209 measurement or that it is finer than the available statistics can support. In such cases, merging or
210 removing the offending bins may be preferable to relying on a projection to handle the resulting
211 constraints. The structural null directions, by contrast, are an unavoidable consequence of measuring
212 multiple distributions from shared events. While their number depends on the binning, as discussed
213 below, they cannot be removed entirely by any choice of bin edges.

214 The structure of the null space of Ω_{struct} depends on how the bin edges for the different blocks
215 overlap. In the simplest case, when blocks do not share variables, every combination of one bin
216 per block is allowed. The resulting null space is $(B - 1)$ -dimensional, corresponding to the $B - 1$
217 independent constraints that require all block totals to be equal. When blocks share one or more
218 variables, the allowed combinations are restricted and the dimension of the null space can be different
219 from $B - 1$. The number of null directions depends on how shared variable bin edges are aligned
220 between blocks. Aligned slice edges decouple the measurement into different sub-problems, adding
221 additional sum rules and increasing $N_{\text{null}}^{\text{struct}}$. On the other hand, misaligned edges couple adjacent
222 distributions, potentially reducing the number of constraints. In the general case, the interplay
223 between shared variables, bin edge alignment, and block topology makes a closed-form expression
224 impractical, but the computation of $\text{rank}(\Omega_{\text{struct}})$ from the bin edge specifications is straightforward.

225 The structural and kinematic components of the null space respond differently to systematic
226 uncertainties. Whether these cause the rank deficiency of C_{stat} to persist, partially resolve, or fully
227 resolve in the total covariance C_{total} depends on the nature of the systematic treatment, which we
228 discuss next.

229 2.4 Effect of systematic uncertainties

230 Previously, we established that the event-sharing structure imposes $N_{\text{null}} = N_{\text{bins}} - \text{rank}(\Omega)$ exact
231 linear constraints on the bin counts, which manifest as a rank deficiency of C_{stat} . Whether the total
232 covariance matrix inherits the rank deficiency depends on which of these null directions receive
233 support from the systematic uncertainties. Now, we examine the mechanisms by which different
234 systematic treatments do or do not populate the null space of C_{stat} .

235 A systematic variation s produces a modified vector of expected events $\mathbf{n}_s$, whose departure
236 from the central-value prediction can be expressed in terms of a shift vector

$$\xi_s = \mathbf{n}_s - \mathbf{n}_{\text{CV}}. \tag{2.11}$$

237 The different contributions to C_{syst} are constructed from the outer product of these shift vectors. A
 238 null direction $\mathbf{x}$ of C_{stat} remains a null direction of C_{total} if and only if $\mathbf{x} \cdot \xi_s = 0$ for every systematic
 239 variation in the uncertainty budget. The question is therefore whether the shift vectors have nonzero
 240 projections onto the structural and kinematic null spaces.

241 Consider first the simplest case, where a systematic s acts by reweighting the reconstructed
 242 event counts directly. Each event e is assigned a modified weight $w_e^{(s)}$ while preserving its bin
 243 assignments. The shift vector can then be written as

$$\xi_s = \sum_e (w_e^{(s)} - w_e) \phi_e. \quad (2.12)$$

244 For any null vector $\mathbf{x}$ which satisfies the null-space condition of Eq. (2.6), the projection of $\mathbf{x}$ onto
 245 the shift vector gives

$$\begin{aligned} \mathbf{x} \cdot \xi_s &= \sum_e (w_e^{(s)} - w_e) (\phi_e \cdot \mathbf{x}) \\ &= \sum_e (w_e^{(s)} - w_e) \sum_{i=1}^{N_{\text{bins}}} \phi_{e,i} x_i \\ &= 0, \end{aligned} \quad (2.13)$$

246 since the inner sum vanishes for every event. Direct reweighting therefore preserves the entire null
 247 space, adding no rank to C_{total} . Flux uncertainties, which modify the neutrino energy spectrum
 248 through per-event weights, are the classic example. Cross-section model variations and hadronic
 249 reinteraction uncertainties, when implemented as per-event reweighting, behave identically.

250 A qualitatively different situation arises when the systematic uncertainties are propagated
 251 through the response matrix rather than changing the event counts per bin directly. In this approach,
 252 adopted as the standard prescription in MicroBooNE for neutrino interaction uncertainties [15], a
 253 modified response matrix Δ'_s , mapping true bins to reconstructed bins under the given variation,
 254 is constructed for each systematic uncertainty. This matrix is obtained either by reweighting the
 255 events used to build the nominal response matrix or from an independent MC sample with varied
 256 simulation parameters. In this case, the predicted reconstructed counts for systematic variation s
 257 are

$$\mathbf{n}_s = \Delta'_s \varphi_{\text{CV}}, \quad (2.14)$$

258 where φ_{CV} is the central-value truth prediction. The shift vector is then

$$\xi_s = (\Delta'_s - \Delta) \varphi_{\text{CV}}. \quad (2.15)$$

259 Since Δ is block-diagonal, the difference $\Delta'_s - \Delta$ acts independently on each block, projecting
 260 the nominal truth vector through block-specific changes in efficiency and migration. The per-event
 261 coupling that makes direct reweighting respect all null directions is absent in this prescription. When
 262 the weight of an event changes, it affects simultaneously every block it populates; the response-
 263 matrix method, in contrast, processes each block independently through its own sub-matrix. The
 264 projection of ξ_s onto any null vector, whether structural or kinematic, therefore involves differences
 265 in how each block responds to the same truth-level prediction, and in general is nonzero. Response-
 266 matrix variations thus break both types of null directions, regardless of how Δ'_s is obtained.

267 A third mechanism applies when systematic uncertainties are evaluated by directly comparing
 268 reconstructed event counts from independent MC samples simulated with nominal and varied
 269 parameters. In this case, the shift vector takes the form

$$\xi_s = \sum_{e'} w_{e'}^{(s)} \phi_{e'}^{(s)} - \sum_e w_e \phi_e, \quad (2.16)$$

270 where the first sum runs over the events in the varied-model sample and the second sum over
 271 the nominal sample. Since each event in both samples populates exactly one bin per block,
 272 their difference projects to zero onto the structural null directions by an analogous argument to
 273 that used for the direct reweighting. Kinematic null directions, however, can be lifted if the
 274 variation scatters events into reconstructed-bin combinations absent in the nominal sample. In that
 275 case, new indicator vectors appear that are not rows of Ω , potentially breaking the corresponding
 276 constraints. Detector systematic uncertainties, which usually require independent samples with
 277 modified simulation parameters that cannot be captured by event reweighting, are the typical
 278 example where this prescription is applied.

279 In practice, whether the eigenvalues of C_{total} along the formerly null directions are exactly
 280 zero or merely smaller than the physical eigenvalues, including these directions in a χ^2 test via
 281 standard matrix inversion produces inflated values with the wrong effective number of degrees
 282 of freedom. Along these directions the measurement carries no statistical information, yet the
 283 systematic model lifts their variance to a small nonzero value; the inverse covariance then weights
 284 them by the reciprocal of that small value. Any residual between data and prediction that projects
 285 onto these directions is amplified, producing a contribution to the χ^2 far in excess of one per degree
 286 of freedom. We make this mechanism precise in Sec. 4.3, where the excess is shown to be a
 287 noncentrality concentrated on the lifted directions. A valid goodness-of-fit test must therefore be
 288 restricted to the subspace where the measurement carries genuine sensitivity, which is the role of
 289 the projection method developed in the following section.

290 3 Range-projected χ^2 test statistic

291 Before constructing the test statistic, we must establish why restricting it to the non-null subspace
 292 discards no physical information. Along a null direction $\mathbf{x}$ of C_{stat} the event-sharing constraint
 293 $\Omega\mathbf{x} = 0$ holds identically. The same selected events enter the bin combinations on both sides,
 294 so the data value along $\mathbf{x}$ is fixed by event conservation rather than measured. Any prediction
 295 forward-folded through the detector response matrix respects the same constraint. The measurement
 296 therefore has no statistical power along $\mathbf{x}$; any residual there, whether a genuine model difference
 297 or an artifact of folding a prediction through the response matrix, carries no statistical meaning,
 298 as the covariance assigns the direction no genuine variance. Discarding these directions removes
 299 no sensitivity, only the ill-defined contributions identified in Sec. 2.4. This holds exactly to the
 300 extent that the event-sharing structure is exact; when sharing is only approximate the null directions
 301 acquire small but nonzero variance, a regime we return to in Sec. 6. Under the exact-sharing
 302 assumption, then, it is safe to formulate the global goodness-of-fit test statistic in the subspace
 303 orthogonal to the constrained directions, which we now construct.

304 The constraints are most naturally identified in reconstructed space, where the combination
 305 matrix is constructed. The non-null subspace can be obtained decomposing $\Omega^T\Omega$. This matrix is

a square $N_{\text{bins}} \times N_{\text{bins}}$ matrix with the same column space and null space as Ω itself, making it a more compact representation of the same information. We can write its eigendecomposition as

$$\Omega^T \Omega = V S V^T, \quad (3.1)$$

where $S = \text{diag}(s_1, \dots, s_{N_{\text{bins}}})$ contains the eigenvalues in decreasing order and V is the orthogonal matrix of eigenvectors. Of these eigenvalues, $r = \text{rank}(\Omega)$ are nonzero. Partitioning $V = [V_r \mid \bar{V}_r]$ into the columns associated with the nonzero and zero eigenvalues, the columns of V_r span the subspace of reconstructed bin space that is not constrained by the event-sharing structure. The orthogonal projector onto this subspace is

$$P_{\text{reco}} = V_r V_r^T. \quad (3.2)$$

Note that P_{reco} is identical to the projector onto the range of C_{stat} as defined in Eq. (2.3), since C_{stat} and $\Omega^T \Omega$ share the same column space by construction. The combination matrix formulation makes explicit the fact that the projector is determined by the measurement geometry.

Cross-section measurements are typically reported in unfolded space, and so it is natural to assume this is the space where goodness-of-fit tests should be evaluated. The blockwise unfolding matrix U , which is block-diagonal and invertible, maps reconstructed-space vectors onto unfolded space. The non-null basis vectors propagate as $W_r = U V_r$. Since U is not orthogonal in general, the columns of W_r are not orthonormal, and the projector onto their span takes the general form

$$P_{\text{true}} = W_r (W_r^T W_r)^{-1} W_r^T. \quad (3.3)$$

Since U is invertible, W_r has the same rank as V_r and the number of constrained directions is preserved. However, as shown next, the ordering of projection and unfolding matters as the commutation relation $U P_{\text{reco}} = P_{\text{true}} U$ generically does not hold. For this reason we choose to define the test statistic in reconstructed space, where the constraints originate.

In practice, the eigenvalues of $\Omega^T \Omega$ that are technically zero take small numerical values. The threshold for identifying these can be set relative to the largest eigenvalue (e.g., $s_i/s_1 < 10^{-10}$). In most cases, a clear spectral gap of several orders of magnitude separates the physical eigenvalues from the numerical noise, making the identification unambiguous.

3.1 Test statistic definition

The goodness-of-fit test is evaluated in reconstructed space, where the residual vector is given by $\delta = \mathbf{d} - \Delta \varphi_{\text{pred}}$, where $\mathbf{d}$ is the background-subtracted data, Δ is the detector response matrix, and φ_{pred} is a truth-level model prediction. The range-projected χ^2 is defined as

$$\chi_{\text{proj}}^2 = \delta^T (P_{\text{reco}} C_{\text{total}} P_{\text{reco}})^+ \delta, \quad (3.4)$$

with the superscript $+$ denoting the Moore–Penrose pseudoinverse.

The pseudoinverse can be avoided entirely by reducing the problem to r dimensions. One can verify that the pseudoinverse of $P_{\text{reco}} C_{\text{total}} P_{\text{reco}}$ takes the closed form

$$(P_{\text{reco}} C_{\text{total}} P_{\text{reco}})^+ = V_r (V_r^T C_{\text{total}} V_r)^{-1} V_r^T. \quad (3.5)$$

Substituting into Eq. (3.4) and defining the reduced residual $\tilde{\delta} = V_r^\top \delta$ and the reduced covariance matrix $\tilde{C}_{\text{total}} = V_r^\top C_{\text{total}} V_r$, the test statistic becomes

$$\chi_{\text{proj}}^2 = \tilde{\delta}^\top \tilde{C}_{\text{total}}^{-1} \tilde{\delta}, \quad (3.6)$$

which involves only a standard inversion of the $r \times r$ matrix $\tilde{C}_{\text{total}}$. Under the null hypothesis, χ_{proj}^2 follows a χ^2 distribution with $r = N_{\text{bins}} - N_{\text{null}}$ degrees of freedom.

Without the projection, the χ^2 test statistic takes the same numerical value whether computed in reconstructed or unfolded space, as the factors of the unfolding matrix cancel identically when applying the additional smearing matrix to the predictions [17]. The introduction of the projector breaks this equivalence when the projection is performed after unfolding. In unfolded-space the residual is defined as $\mathbf{r} = \hat{\phi} - A_C \phi_{\text{pred}}$, where $\hat{\phi} = U\mathbf{d}$ is the unfolded background-subtracted data and $A_C = U\Delta$ is the additional smearing matrix. Using the propagated basis $W_r = U V_r$, one can construct an analogous test statistic in unfolded space as

$$\chi_{\text{unf}}^2 = \tilde{\mathbf{r}}^\top \tilde{V}_{\text{total}}^{-1} \tilde{\mathbf{r}}, \quad (3.7)$$

where $V_{\text{total}} = U C_{\text{total}} U^\top$ is the total covariance matrix in unfolded space, and $\tilde{\mathbf{r}} = W_r^\top \mathbf{r}$ and $\tilde{V}_{\text{total}} = W_r^\top V_{\text{total}} W_r$ are the reduced residual and reduced covariance matrix in unfolded space. While this is a valid test statistic on its own, it is not numerically equal to Eq. (3.6). Expressing it in terms of reconstructed-space quantities yields

$$\chi_{\text{unf}}^2 = \delta^\top U^\top U V_r (V_r^\top U^\top U C_{\text{total}} U^\top U V_r)^{-1} V_r^\top U^\top U \delta, \quad (3.8)$$

which reduces to Eq. (3.6) only when $U^\top U = I$, i.e. when the unfolding matrix is orthogonal.

The χ^2 equivalence is restored if the projection is applied before unfolding. Projecting the reconstructed-space residual and covariance first, then propagating through the unfolding matrix, gives

$$\begin{aligned} (U P_{\text{reco}} \delta)^\top (U P_{\text{reco}} C_{\text{total}} P_{\text{reco}} U^\top)^+ (U P_{\text{reco}} \delta) &= \tilde{\delta}^\top W_r^\top (W_r \tilde{C}_{\text{total}} W_r^\top)^+ W_r \tilde{\delta} \\ &= \tilde{\delta}^\top G (G \tilde{C}_{\text{total}} G)^{-1} G \tilde{\delta} \\ &= \chi_{\text{proj}}^2, \end{aligned} \quad (3.9)$$

where the symmetric matrix $G = W_r^\top W_r$ acts as the unfolding matrix of the reduced system, and cancels similarly to the unprojected case. In the approach of Eq. (3.8), the full residual, including the null-space components, is first rotated by U into directions that are no longer orthogonal to the range. The unfolded-space projector then picks up this null-space contamination. Projecting first in reconstructed space avoids this by removing the null directions before unfolding can mix them with the physical subspace.

For this reason, Eq. (3.6) is the recommended definition of the range-projected χ^2 .

3.2 Statistical validity

Under the null hypothesis, the residual δ is drawn from a multivariate Gaussian distribution with mean zero and covariance C_{total} . The reduced residual $\tilde{\delta} = V_r^\top \delta$ is a linear transformation of a Gaussian vector, therefore Gaussian itself with mean zero and covariance $\tilde{C}_{\text{total}} = V_r^\top C_{\text{total}} V_r$.

366 Provided that $\tilde{C}_{\text{total}}$ is positive definite, standard theory of Gaussian quadratic forms guarantees that
 367 $\tilde{\delta}^T \tilde{C}_{\text{total}}^{-1} \tilde{\delta} \sim \chi^2(r)$. The positive definiteness of $\tilde{C}_{\text{total}}$ requires C_{total} to have nonzero variance along
 368 every direction in the range of the projector P_{reco} , which is generically satisfied. This can be verified
 369 by checking that all eigenvalues of $\tilde{C}_{\text{total}}$ are strictly positive.

370 The validity of the range-projected χ^2 can also be understood from the general theory of
 371 quadratic forms of correlated normal variables. Theorem 9.2.2 of Ref. [22] establishes that if
 372 $\mathbf{Y} \sim \mathcal{N}(0, \Sigma)$ with $\text{rank}(\Sigma) = k$, then $\mathbf{Y}^T \Sigma^+ \mathbf{Y} \sim \chi^2(k)$ for any choice of generalized inverse Σ^+ .
 373 Assuming the null hypothesis, the projected residual $P_{\text{reco}} \delta$ is normally distributed with covariance
 374 $P_{\text{reco}} C_{\text{total}} P_{\text{reco}}$, which has rank r by construction. The formulation of Eq. (3.4) with the Moore–
 375 Penrose pseudoinverse is a specific instance of this general result. The key distinction from purely
 376 numerical approaches to handling singular covariance matrices is that the rank is not determined
 377 by an eigenvalue threshold applied to the covariance, but is predicted exactly via the combination
 378 matrix Ω from the structure of the measurement.

379 3.3 Relation to alternative approaches

380 The range-projected χ^2 is, as a statistical object, an instance of the standard treatment of quadratic
 381 forms with singular covariance matrices described in Sec. 3.2; a generalized inverse restricted to the
 382 range of the covariance, distributed as a χ^2 with degrees of freedom equal to the rank [22]. Dropping
 383 zero-variance directions from a χ^2 is common practice in fits with degenerate covariance matrices.
 384 The contribution of this work is not the statistic itself but the recognition that the rank deficiency
 385 arises specifically and predictably from the shared-event structure, together with a prescription to
 386 compute its dimension a priori from the binning geometry. We discuss here how this differs from
 387 related approaches.

388 The most direct alternative is to regularize the covariance matrix numerically, for instance
 389 by performing a singular value decomposition and discarding eigenvalues below a fixed threshold
 390 relative to the largest. When the rank deficiency is exact this is unambiguous, but the systematic
 391 uncertainties of a realistic measurement lift the null eigenvalues to small but nonzero values, and
 392 these lifted directions need not be separated from the genuine physical eigenvalues by a clean gap.
 393 A threshold must then be chosen to sit between the lifted null directions and the smallest physical
 394 ones, with no guarantee that such a value exists. Setting it too low retains the inflating directions
 395 discussed next in Sec. 4.3, while setting it too high discards directions carrying real information.
 396 The combination-matrix approach avoids this choice. In this case the rank is fixed by Ω from
 397 the measurement geometry, independently of the systematics-contaminated spectrum of C_{total} . A
 398 threshold is still used to identify the null space, but it is applied to $\Omega^T \Omega$, where the gap is clean by
 399 construction, rather than to C_{total} where systematics have filled it in.

400 The projection is distinct from, and complementary to, the regularization performed in unfold-
 401 ing. Methods such as Wiener-SVD [17] address the ill-posedness of inverting the response matrix,
 402 producing the additional smearing matrix A_C that is applied to truth-level predictions before com-
 403 parison. They do not address the event-sharing rank deficiency, which is a property of the statistical
 404 covariance and survives into unfolded space regardless of the unfolding method. The two operate
 405 on different sources of degeneracy and are used together. The event generator study of Sec. 5
 406 unfolds each block with Wiener-SVD and applies the range projection on top.

A more complete alternative is to extract the cross section directly in a binned maximum likelihood fit, in which the truth-level bins are free parameters of interest and the systematic uncertainties enter as nuisance parameters with prior constraints (see, e.g., Ref. [23]). The likelihood is evaluated in reconstructed space, where the inter-distribution correlations are carried by the shared events and the common nuisance parameters of the forward model; it never forms an inverted covariance over the bin counts, so the constrained directions are absorbed automatically and the goodness of fit can be assessed without confronting the rank deficiency at all. This approach is also more robust to the imperfect-sharing scenarios discussed in Sec. 6, where the constraints are only approximate. Its requirements are correspondingly greater: it needs the full forward model and nuisance parameterization, which are generally available only within the originating group. In this regard, the range-projected χ^2 requires far less information.

For the group producing a blockwise-unfolded measurement it is the natural global goodness-of-fit statistic, computed directly from the covariance and response matrices already used in the extraction, and evaluated in reconstructed space as recommended in Sec. 3.1. It applies equally to an already-published measurement, as performed by generator comparison and tuning frameworks [24]. A standard data release reporting the unfolded data, unfolded covariance, and the block-diagonal unfolding matrix is sufficient to recover the reconstructed-space data and covariance by inversion; the reconstructed-space test then additionally requires the detector response matrix to forward-fold a model prediction. Where the response matrix is not available, the projection can instead be performed in unfolded space using the additional smearing matrix A_C , which is an increasingly standard release product. This unfolded-space test is less exact, as the non-orthogonal unfolding mixes a residual null-space contamination into the physical subspace, but it still removes the bulk of the inflation. The two approaches are complementary: profiling is more general when the complete model is available, while the projection provides a lightweight, a priori test from published results.

4 Toy example

The formalism developed previously is now illustrated with a toy model that captures the essential features of a multi-distribution neutrino cross-section measurement: a two-dimensional underlying physics process, a simple detector model, and a multi-block binning scheme in which events are shared across distributions. The toy is designed to be minimal but non-trivial. A three-block structure produces a rank-deficient statistical covariance matrix, and a confidence-region coverage study demonstrates the consequences of ignoring or correctly treating the resulting null directions.

4.1 Model and measurement setup

The toy model mimics a muon neutrino charged-current quasi-elastic-like interaction, producing a muon characterized by its momentum p_μ and scattering angle θ_μ . The joint truth-level differential cross section is parametrized as

$$f(p_\mu, \cos \theta_\mu) = p_\mu^k e^{-k p_\mu/p_0} e^{\kappa \cos \theta_\mu} (1 + \alpha p_\mu \cos \theta_\mu), \quad (4.1)$$

where the four parameters (k, p_0, κ, α) control the shape of the momentum spectrum, the level of forward peaking, and the strength of the momentum–angle correlation, respectively. A central-

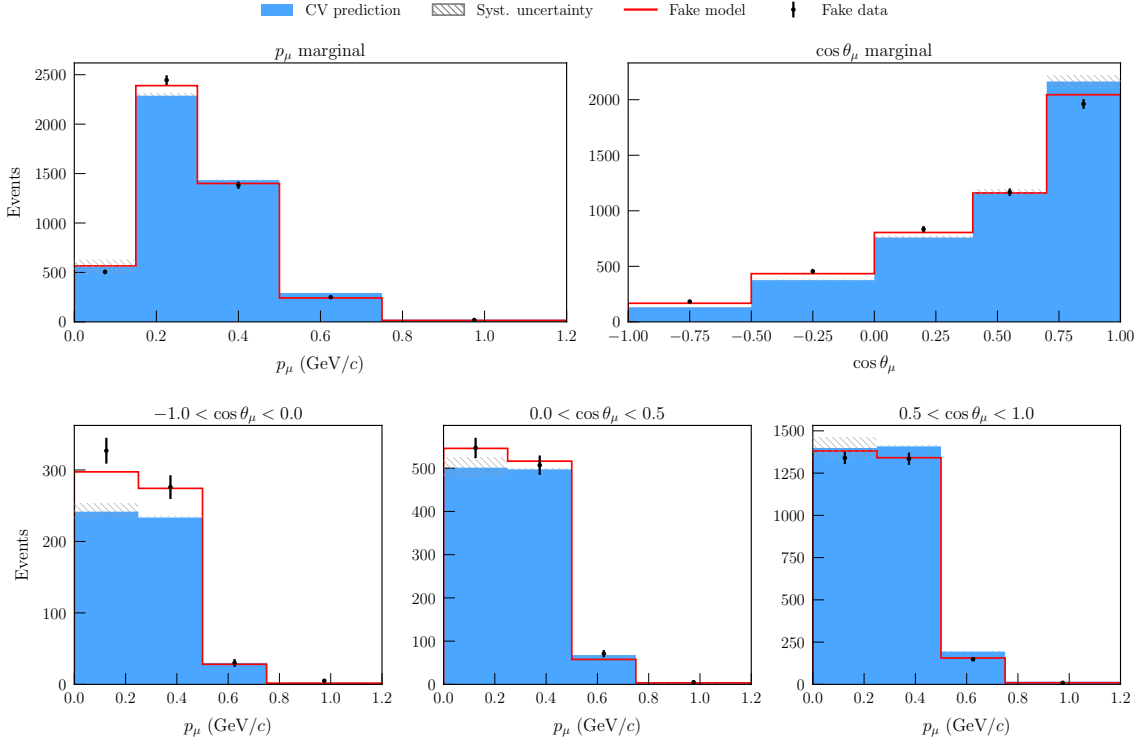


Figure 1. Predicted event distributions in the toy analysis blocks for the CV model (filled histograms) and the fake model (solid lines), together with a pseudo-data realization (points with statistical error bars). The shaded bands indicate the systematic uncertainty on the CV prediction. The top row shows the p_μ and $\cos \theta_\mu$ marginals; the bottom row shows the three $\cos \theta_\mu$ slices of the 2D block, each binned in p_μ .

value (CV) model with $(k, p_0, \kappa, \alpha) = (3.0, 0.2, 2.0, 0.3)$ is used to build the response matrix and estimate the covariance, while a “fake” model with $(k, p_0, \kappa, \alpha) = (3.3, 0.2, 1.8, 0.35)$ serves as the true underlying distribution from which pseudo-data are drawn. Truth events are generated via rejection sampling over $p_\mu \in [0.0, 1.2]$ GeV/c and $\cos \theta_\mu \in [-1.0, 1.0]$.

A simple detector model introduces two different effects. Momentum is smeared multiplicatively, $p_{\text{reco}} = p_{\text{true}}(1 + \delta_p)$ with $\delta_p \sim \mathcal{N}(0, \sigma_p)$ and $\sigma_p = 0.05$, while the angle is smeared additively, $\theta_{\text{reco}} = \theta_{\text{true}} + \delta_\theta$ with $\delta_\theta \sim \mathcal{N}(0, \sigma_\theta)$ and $\sigma_\theta = 1^\circ$. A sigmoid efficiency function $\varepsilon(p_{\text{reco}}) = [1 + e^{-(p_{\text{reco}} - p_{\text{thr}})/w}]^{-1}$ with threshold $p_{\text{thr}} = 0.10$ GeV/c and width $w = 0.02$ GeV/c models the turn-on of the event acceptance at low momentum.

The analysis uses three bin blocks in reconstructed space: a p_μ marginal (5 bins), a $\cos \theta_\mu$ marginal (5 bins), and a two-dimensional distribution sliced in $\cos \theta_\mu$ (3 slices, each with 4 bins in p_μ), for a total of $N_{\text{bins}} = 22$. The bin edges of the 2D block do not align exactly with those of the 1D marginals, which is the usual situation in real analyses. A CV sample of $N_{\text{MC}} = 500,000$ truth events is generated, processed through the detector model, and used to construct the response matrix Δ . Pseudo-data samples are generated by drawing $N_{\text{data}} = 5,000$ events from the fake model.

Systematic uncertainties are estimated following the response-matrix approach described in Sec. 2.4. Interaction-model variations are obtained by reweighting the CV sample to six alternative model configurations, each producing a varied response matrix Δ_s . The shift vector $\xi_s = (\Delta_s -$

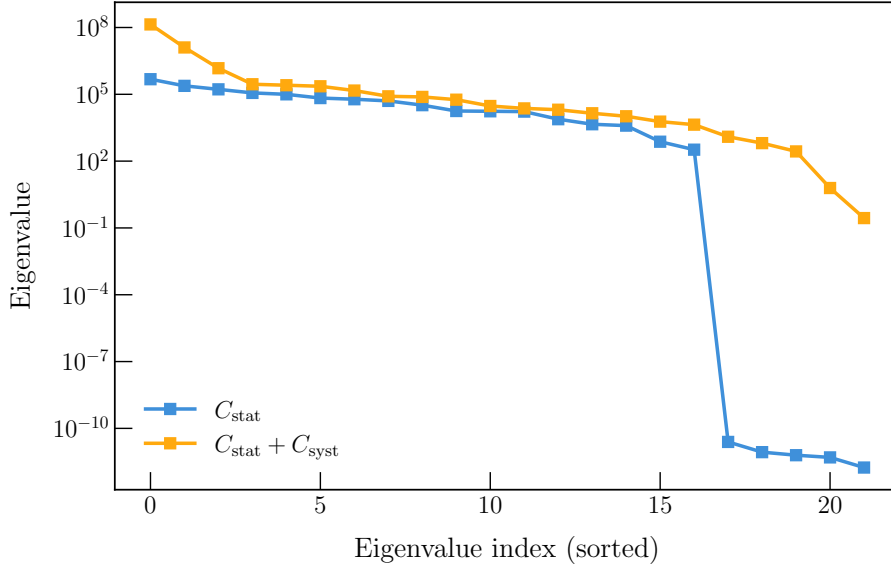


Figure 2. Eigenvalue spectra of the statistical covariance matrix C_{stat} (blue circles) and the total covariance matrix $C_{\text{total}} = C_{\text{stat}} + C_{\text{syst}}$ (orange squares), for the toy analysis, sorted in descending order.

463 Δ) φ captures the effect on the predicted reconstructed-space distribution. Detector variations are
 464 obtained by generating eight independent samples with perturbed detector parameters (σ_p , σ_θ , p_{thr} ,
 465 and w), each producing a new response matrix from which a shift vector is computed similarly. The
 466 total systematic covariance matrix is $C_{\text{syst}} = N_{\text{var}}^{-1} \sum_s \xi_s \xi_s^T$, scaled to data statistics. The statistical
 467 covariance is computed using Eq. (2.2), assuming unit event weights, which captures the inter-block
 468 correlations from event sharing.

469 Figure 1 shows the predicted event distributions for the CV and fake models in all three blocks
 470 of the toy analysis, together with a pseudo-data sample distribution and the systematic uncertainty
 471 bands.

472 4.2 Null-space structure

473 The structural combination matrix Ω_{struct} is constructed from the binning configuration of the three
 474 blocks, following the prescription from Sec. 2.3. For this bin scheme, Ω_{struct} is rank 17, yielding
 475 5 structural null directions. The combination matrix Ω computed from the MC event list also has
 476 rank 17, confirming that all 5 null directions are structural. No additional kinematic constraints
 477 arise for this choice of physics model and binning.

478 The eigenvalue spectrum of C_{stat} confirms this. It shows 5 eigenvalues that are numerically zero,
 479 separated from the 17 nonzero eigenvalues by a spectral gap of several orders of magnitude. Adding
 480 systematic uncertainties constructed via the response-matrix approach lifts the 5 null eigenvalues,
 481 making $C_{\text{total}} = C_{\text{stat}} + C_{\text{syst}}$ formally full rank. However, as shown in Fig. 2, the lifted eigenvalues
 482 remain considerably smaller than the bulk of the spectrum. Including them in the test statistic
 483 introduces poorly calibrated contributions to the χ^2 , as demonstrated next.

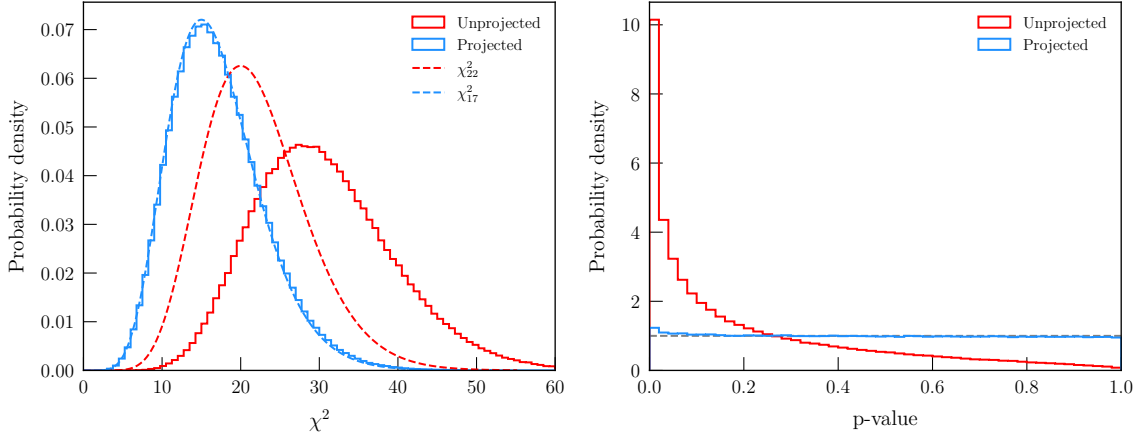


Figure 3. Goodness-of-fit validation for toy experiments generated at the true hypothesis. Left: distributions of the unprojected (red) and range-projected (blue) χ^2 statistics, compared with the expected $\chi^2(22)$ and $\chi^2(17)$ distributions (dashed curves). Right: corresponding p-value distributions, with the false rejection rates at the 5% significance level indicated.

4.3 Goodness-of-fit validation

The calibration of the projected and unprojected χ^2 test statistics is assessed with a toy Monte Carlo exercise at the true hypothesis point. A high-statistics simulation of the fake model is used to estimate the event distribution on a fine reconstructed-space grid, defined by the union of all bin edges in p_μ and $\cos \theta_\mu$ across the three blocks, corresponding to the common refinement described in Sec. 2.3. Each toy experiment draws a Poisson-fluctuated total event count and distributes them across the fine grid via multinomial sampling. These events are then projected onto the three analysis blocks, preserving the inter-block correlations from event sharing. Systematic fluctuations are added as correlated Gaussian throws from the Cholesky decomposition of C_{sys} . The χ^2 is evaluated in reconstructed space, comparing each toy to the forward-folded prediction at the true hypothesis, for both the unprojected statistic, using the direct inverse of C_{total} , and the range-projected statistic of Eq. (3.4).

Figure 3 shows the results from 10^6 toy experiments. The projected χ^2 statistic closely follows the expected $\chi^2(17)$ distribution, with a mean of 17.23 and a standard deviation of 6.01, close to the expected values of 17 and $\sqrt{2 \times 17} \approx 5.83$. In contrast, the unprojected statistic is shifted to significantly higher values with a mean of 30.59, and a standard deviation of 8.96, far above the expectation for a $\chi^2(22)$ distribution of 22 and $\sqrt{2 \times 22} \approx 6.63$. The right panel shows the corresponding p-value distributions. The projected statistic yields an approximately uniform distribution, with a false rejection rate of 5.7% at the 5% significance level. The unprojected statistic produces a false rejection rate of 32.4%, more than six times the expected value. An analyzer using the unprojected statistic for goodness-of-fit testing would falsely reject the correct model roughly one third of the time.

Writing the residual vector at the true hypothesis point as $\delta = \delta_{\text{rand}} + \mathbf{b}$, where δ_{rand} has covariance C_{total} and $\mathbf{b} \equiv E[\delta]$ is the deterministic data–prediction residual (estimated as the mean

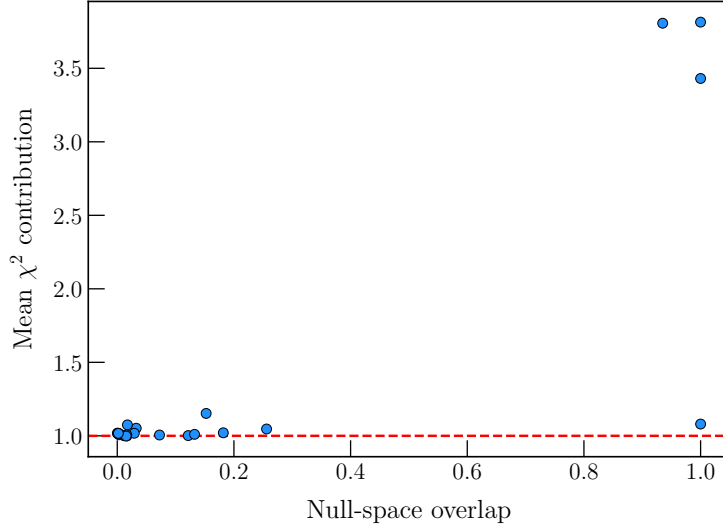


Figure 4. Mean contribution to the unprojected χ^2 from each eigendirection of C_{total} , averaged over the toy ensemble, versus the null-space overlap O_j from Eq. (4.4).

over the toy throws), and diagonalizing $C_{\text{total}} = \sum_j \lambda_j \mathbf{v}_j \mathbf{v}_j^T$, the unprojected statistic decomposes as

$$\chi_{\text{unproj}}^2 = \sum_j \frac{(\boldsymbol{\delta} \cdot \mathbf{v}_j)^2}{\lambda_j}, \quad (4.2)$$

with the expected value of each of these contributions given by

$$\mathbb{E} [(\boldsymbol{\delta} \cdot \mathbf{v}_j)^2 / \lambda_j] = 1 + \frac{(\mathbf{b} \cdot \mathbf{v}_j)^2}{\lambda_j}. \quad (4.3)$$

Each eigendirection contributes as a noncentral $\chi^2(1)$ with noncentrality parameter $\psi_j = (\mathbf{b} \cdot \mathbf{v}_j)^2 / \lambda_j$. On the directions overlapping the null space of C_{stat} the eigenvalue λ_j is the small lifted systematic variance, so any residual projecting onto them is amplified into a contribution far exceeding one.

To identify which directions drive the excess in the unprojected statistic, we quantify how much each eigenvector of C_{total} overlaps the null space of C_{stat} . Using the projector onto the non-null subspace P_{reco} from Eq. (3.2), we define the overlap coefficient as

$$O_j = \mathbf{v}_j^T (I - P_{\text{reco}}) \mathbf{v}_j, \quad (4.4)$$

which equals one for a direction lying entirely in the null space and zero for one in the range. Figure 4 shows the mean contributions to the χ^2 test statistic across the toy ensemble against O_j . The 17 non-null directions cluster around $O_j \approx 0$ with contributions consistent with unity. Across the ensemble the random part of the residual is correctly calibrated along every eigendirection, with the empirical variance matching λ_j to better than 1% on both the lifted and the range directions; the excess is therefore entirely noncentrality. The directions carrying the largest contributions are all those with substantial null-space overlap, where the small lifted eigenvalues amplify any residual projecting onto them. Summing over all directions, the noncentrality reproduces the observed excess, $\sum_j \psi_j \approx 8.6$, accounting for the unprojected mean of 30.6.

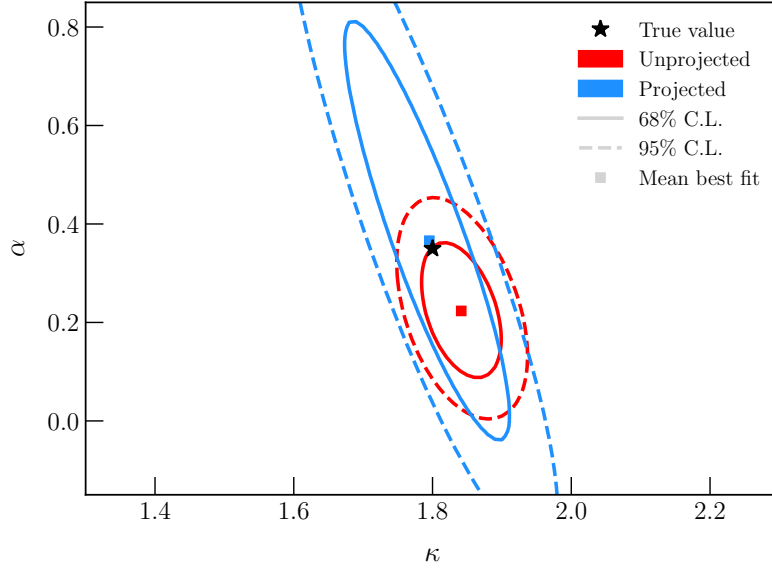


Figure 5. Confidence regions in the (κ, α) plane from the Asimov dataset, showing the 68% and 95% CL contours for the range-projected (blue) and unprojected (red) test statistics. The star marks the true parameter values; the squares indicate the mean best-fit points across 10^6 toys.

4.4 Parameter estimation

The impact of using the projected test statistic on parameter estimation is assessed with a two-dimensional confidence-region study. We scan the two parameters (κ, α) that control the angular shape and the momentum–angle correlation, holding the momentum–spectrum parameters (k, p_0) fixed at their fake-model values, as these are well constrained by the p_μ marginal and largely decoupled from the null-space directions of interest. At each grid point the truth-level prediction is forward-folded through the nominal response matrix and compared to the pseudo-data, and the profiled test statistic $\Delta\chi^2 = \chi^2(\kappa, \alpha) - \chi^2_{\min}$ is evaluated for both the unprojected and projected cases. Confidence regions are defined by the standard thresholds $\Delta\chi^2 < F^{-1}(\text{CL})$ for n scanned (or profiled) parameters, and their frequentist coverage is then measured directly from the toy ensemble.

Figure 5 shows the 68% and 95% confidence level contours from the Asimov dataset for both test statistics. The projected contours are correctly centered on the true parameter values, while the unprojected contours are shifted, with the 68% region excluding the true point. The unprojected best fit values are biased, particularly in α . The mean best-fit value across toys is $\hat{\alpha} = 0.22$, far from the true value of 0.35, while the projected statistic recovers $\hat{\alpha} = 0.37$. The bias in κ is milder ($\hat{\kappa} = 1.84$ for unprojected, $\hat{\kappa} = 1.80$ for projected, compared to the true 1.80). This indicates that α has greater overlap with the null directions.

The frequentist coverage confirms these observations. At the 68.3% confidence level, the projected statistic achieves nominal coverage in both the joint two-dimensional region (68.2%) and the one-dimensional profiled intervals (68.1% for κ , 67.9% for α). The unprojected statistic suffers severe undercoverage: 36.5% in the joint region, 44.6% for κ , and 34.1% for α . The undercoverage is worst for α , consistent with its stronger sensitivity to the null-space contamination. The projected contours are larger than the unprojected ones, reflecting the genuine sensitivity of the measurement

once the ill-constrained directions are properly removed.

5 Event generator study

The toy study presented in the previous section validates the usage of the range-projected χ^2 test statistic in a controlled setting, with an analytic cross-section model and known input parameters. Now, we demonstrate that the same pathology arises, and the same fix applies, when working with a realistic measurement setup. In this case, we use a real neutrino flux, neutrino event generators, cross-section systematic uncertainties, and unfolding strategy. The latter is a key addition over the toy study, as it allows us to verify numerically the equivalence between the results in reconstructed and unfolded space, established in Sec. 3.1. In addition, we show that the projected test statistic retains its discriminating power between generator predictions, correctly rejecting the wrong model while the unprojected statistic does not.

5.1 Model and measurement setup

The central-value model is built from GENIE v3.6.2 [25] using the AR23_20i_00_000 tune on an ^{40}Ar target. The fake-data model, which plays the role of the unknown true cross section, is NuWro v25.11 [26] on the same target. Both generators are run with the SBND BNB ν_μ flux over the energy range 0–10 GeV. Since both predictions use the same flux, flux systematic uncertainties cancel in this comparison and are not considered; a measurement comparing against an external flux prediction would include them as an additional systematic.

Events are selected using a topological $\text{CC}1\mu1p0\pi$ definition applied identically to both generators. We require exactly one muon, exactly one proton above 40 MeV kinetic energy, and no charged or neutral mesons above 50 MeV in the final state. This topological selection captures CCQE, CCMEC, and CCRES events where the pion was absorbed in final-state interactions.

The detector model is the same as in the toy study. For each candidate event, we apply a Gaussian momentum and angular smearing ($\sigma_p = 5\%$, $\sigma_\theta = 1^\circ$), and a sigmoid efficiency threshold at $p_{\text{thr}} = 100$ MeV with a width $w = 20$ MeV to the primary muon.

The analysis binning follows the same three-block scheme as the toy study: a p_μ marginal, a $\cos\theta_\mu$ marginal, and a two-dimensional block sliced in $\cos\theta_\mu$. In the present sample, however, the highest-momentum bin of the most backward $\cos\theta_\mu$ slice of the two-dimensional block receives no selected events, as the $\text{CC}1\mu1p0\pi$ topology has negligible acceptance for high-momentum backward-going muons. This empty bin is merged with its neighbor, leaving $5 + 5 + (3 + 4 + 4) = 21$ bins across the three blocks.

One million GENIE events are generated and used to construct the response matrix and estimate the systematic uncertainties. One million NuWro events are generated separately; 5,000 of these are drawn as pseudo-data, with the remainder serving as the high-statistics truth prediction.

5.2 Systematic uncertainties and unfolding

Interaction-model systematic uncertainties are estimated using the GENIE Reweight framework [25], which provides per-event weights at $\pm 1\sigma$ for each systematic knob. Fifteen knobs are varied, spanning the dominant sources of uncertainty for the $\text{CC}1\mu1p0\pi$ topology: four z -expansion coefficients for the CCQE axial form factor [27], RPA screening and Pauli blocking corrections [28],

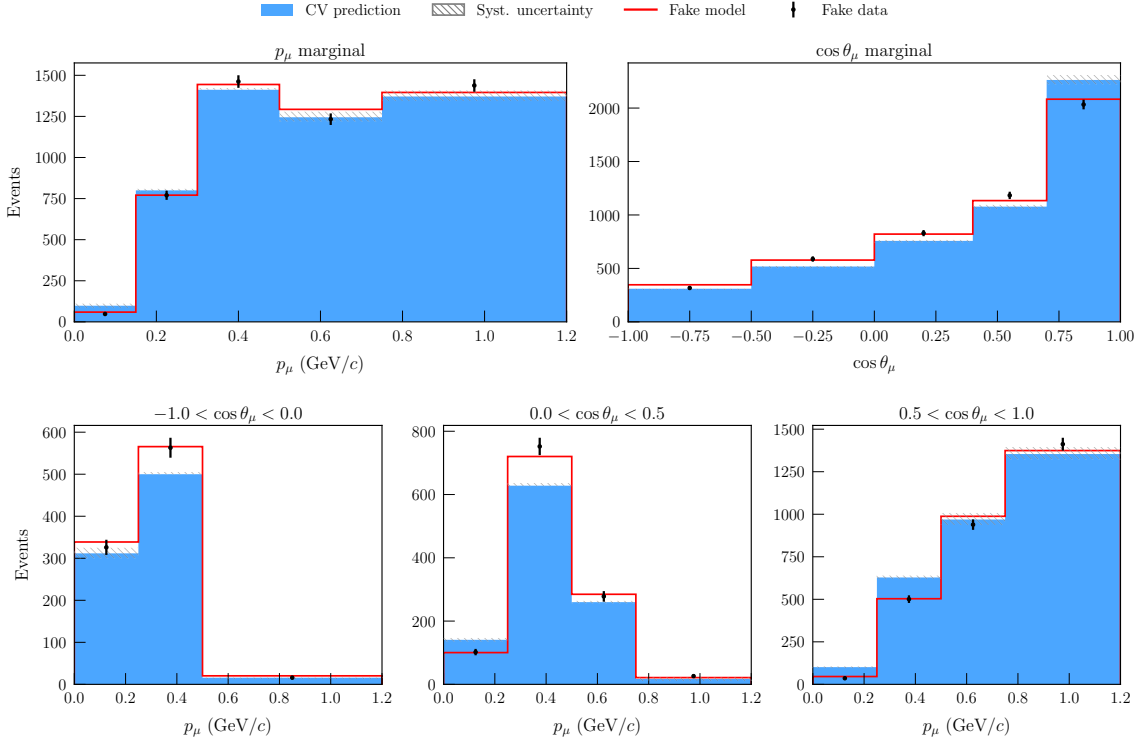


Figure 6. Reconstructed-space event distributions for the event generator study. The GENIE CV prediction (filled histograms) and systematic uncertainty bands (shaded) are compared to the NuWro prediction (red line) and a 5,000-event NuWro pseudo-data realization (points with statistical error bars). The top row shows the p_μ and $\cos \theta_\mu$ marginals; the bottom row shows the three $\cos \theta_\mu$ slices of the 2D block.

the resonance axial and vector masses and overall normalization [29], the CCMEC normalization, two non-resonant single-pion background parameters, and three final-state interaction parameters controlling the pion mean free path, absorption fraction, and charge-exchange fraction [30]. Detector uncertainties are estimated similarly, using $\pm 1\sigma$ variations of the four detector parameters (σ_p , σ_θ , p_{thr} , w). The systematic covariance matrices are constructed via the response-matrix approach described in Sec. 2.4.

Each of the three distribution blocks is unfolded independently using the Wiener-SVD approach [17], with the response matrix and total data covariance computed block-by-block. The per-block unfolding matrices are assembled into the block-diagonal matrix U , and the additional smearing matrix A_C is also constructed from the per-block results. The full covariance is propagated through U as $V_{\text{total}} = U C_{\text{total}} U^T$, preserving the cross-block correlations from event sharing. Truth-level predictions are smeared with A_C before comparison to the unfolded data.

5.3 Results

The structural combination matrix Ω_{struct} , built from the three-block binning scheme, has rank 17, yielding $N_{\text{null}}^{\text{struct}} = 21 - 17 = 4$ structural null directions. The full combination matrix Ω , computed from the GENIE event list, reveals one additional kinematic null direction, bringing the total to $N_{\text{null}} = 5$. This is distinct from the bin merging mentioned in Sec. 5.1. In that case we merged a

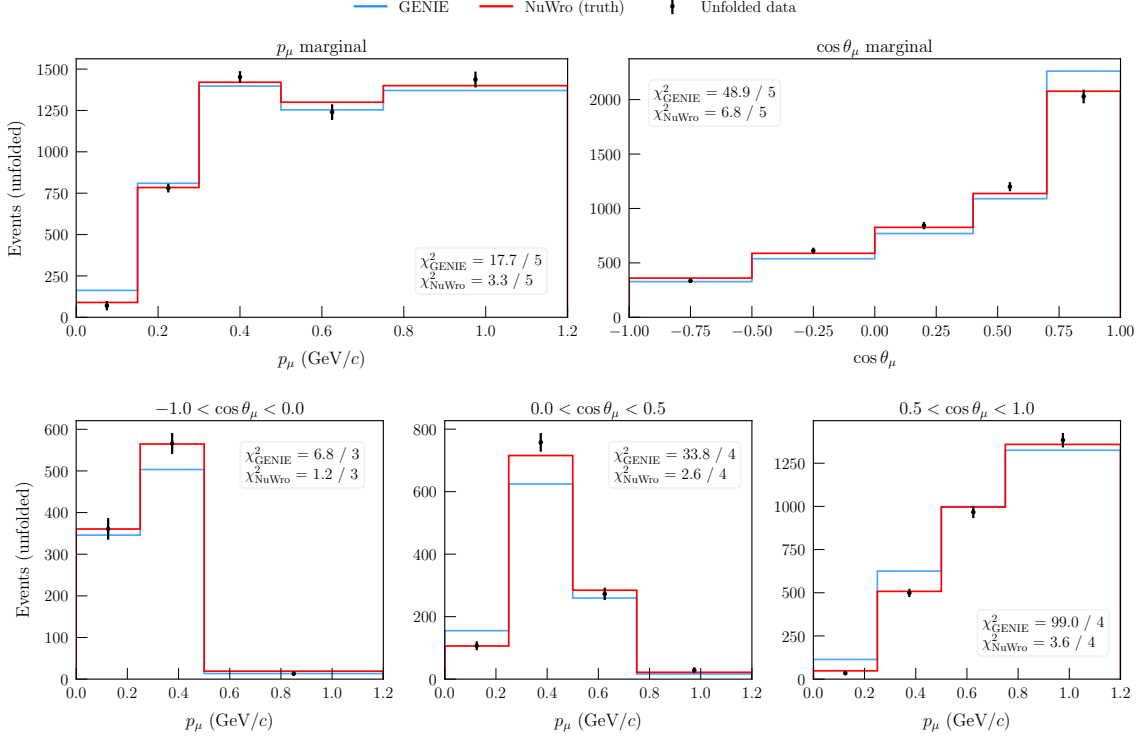


Figure 7. Unfolded event distributions for the event generator study. The GENIE (blue) and NuWro (red) truth-level predictions are smeared through the additional smearing matrix A_C for comparison with the unfolded fake data (points with propagated uncertainties). Per-block χ^2 values are annotated in each panel.

single bin of zero occupancy, whereas the kinematic null is a linear constraint among the populated bins. It arises because two cells of the common refinement corresponding to high-momentum backward-going muons are not populated by any selected events, a region where the CCQE-like cross section is strongly suppressed. The marginal bins themselves remain populated; only the fine-grained intersection of the high-momentum and backward-angle regions is empty. Their absence removes one independent bin combination from Ω , reducing its rank to 16. The projected test statistic therefore has $r = 16$ degrees of freedom.

Figure 6 shows the reconstructed-space event distributions in all three blocks. The GENIE CV prediction and its systematic uncertainty band are compared to the NuWro fake-data realization and the NuWro prediction. The GENIE systematic band does not fully cover the NuWro prediction in several bins, reflecting genuine model differences that the χ^2 test should be able to identify.

Figure 7 shows the corresponding unfolded distributions, with the GENIE and NuWro truth-level predictions smeared through the additional smearing matrix A_C . Per-block χ^2 values, computed using only the block-diagonal covariance, are annotated in each panel. These are well-behaved for the NuWro comparison, confirming that the unfolding correctly recovers the true distribution at the per-block level. The GENIE comparison shows mild tension in some blocks.

Table 1 summarizes the global χ^2 results. The unprojected statistic yields 121.9/21 for the NuWro prediction. This results in a disproportionate rejection of the correct model, entirely driven by the null-space contamination. The projected statistic gives 15.7/16, consistent with the

Table 1. Global χ^2 values for the NuWro fake data compared to the GENIE and NuWro truth-level predictions. The projected test statistic has $N_{\text{bins}} - N_{\text{null}} = 16$ degrees of freedom; the unprojected has $N_{\text{bins}} = 21$.

Test statistic	GENIE	NuWro
Unprojected χ^2/ndof	252.4 / 21	121.9 / 21
Projected χ^2/ndof	154.4 / 16	15.7 / 16

624 expected $\chi^2(16)$ distribution and confirming that the method is correctly calibrated. This was
 625 further validated with 10^5 pseudo-experiments, which confirmed that the projected χ^2 distribution
 626 follows $\chi^2(16)$ while the unprojected distribution does not follow $\chi^2(21)$, consistent with the toy
 627 study results of Sec. 4.3.

628 The projected statistic retains full discriminating power against the wrong model. The com-
 629 parison to GENIE yields 154.4/16, a strong rejection indicating that the GENIE uncertainty budget
 630 cannot accommodate the NuWro prediction. By contrast, the unprojected statistic gives 252.4/21
 631 for GENIE, also a strong rejection but indistinguishable in practice from its 121.9/21 for NuWro.
 632 Without the projection, both the correct and incorrect models are rejected, and the test provides no
 633 useful discrimination.

634 Finally, the projected χ^2 was verified to give identical numerical values whether computed in
 635 reconstructed or unfolded space, for both the GENIE and NuWro predictions. This confirms the
 636 equivalence established in Sec. 3.1 and validates the practical implementation of the projection in
 637 the presence of unfolding.

638 6 Limitations and scope

639 The proposed method rests on the assumption that the event-sharing structure is exact. Every
 640 selected event populates exactly one bin in each distribution, so the constraints $\Omega \mathbf{x} = 0$ hold
 641 identically and the null space of C_{stat} , built according to Eq. (2.2), coincides with that of Ω . Under
 642 this assumption the measurement has no statistical power along a null direction, so projecting it out
 643 discards no statistical information, as argued in Sec. 3.

644 In real measurements, this assumption is sometimes only approximate. Distributions extracted
 645 from a common sample may nonetheless use different phase-space limits, apply distribution-
 646 dependent background subtractions, or define their signal components differently, so that the popu-
 647 lations entering different blocks are not strictly identical. Consider the simplest constraint that two
 648 block totals be equal, encoded by the direction $\mathbf{x}$ with entries +1 on the bins of one block and -1
 649 on those of another. Its statistical variance is given by

$$\mathbf{x}^T C_{\text{stat}} \mathbf{x} = \sum_e w_e^2 (\phi_e \cdot \mathbf{x})^2 = \sum_{e \notin \text{both blocks}} w_e^2, \quad (6.1)$$

650 which vanishes only when every event is shared. When a fraction ϵ of events is unshared, the
 651 variance is of order ϵN . In this case, the corresponding eigenvalue is lifted from zero into the body
 652 of the spectrum by the statistical model itself, before any systematic uncertainty is added. This is

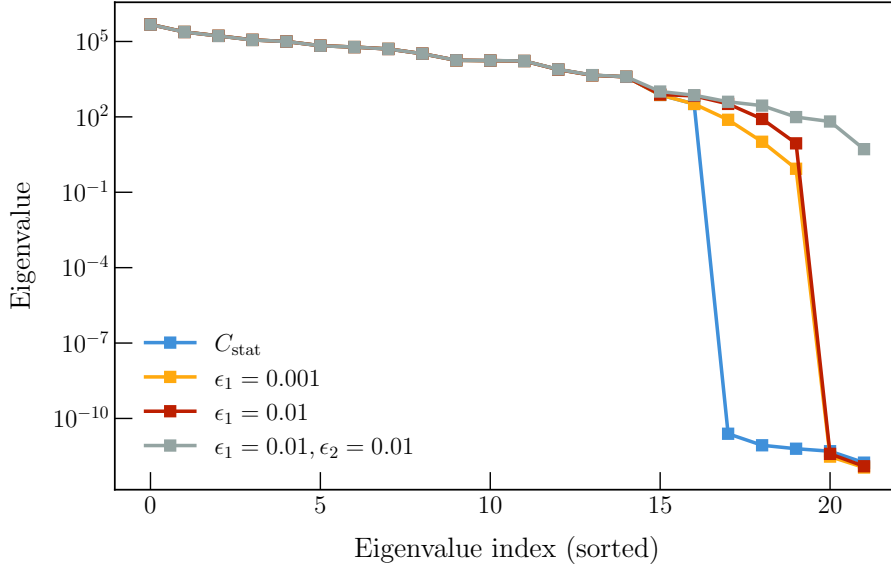


Figure 8. Eigenvalue spectra of the statistical covariance matrix C_{stat} for the toy analysis, sorted in descending order, as events are removed at random from one or two blocks, breaking the exact event-sharing structure. Fractions ϵ_1 and ϵ_2 of events are dropped from the p_μ and $\cos \theta_\mu$ marginal blocks, respectively.

a qualitatively different situation from the inflation of Sec. 4.3. There the variance along some directions was exactly zero in C_{stat} and lifted only by the systematic model to a value unrelated to any statistical power, so dividing a residual by it produced an unphysical contribution. A direction carrying a genuine variance ϵN from real event mismatch is not pathological, as the test weights it by its true variance and remains calibrated. The risk of imperfect sharing is therefore not renewed inflation but over-projection, i.e. discarding a direction that now carries a small but real amount of information.

The risk is confined to the a priori structural prescription, and it can be checked for. Three objects encode the null space under exact sharing, and they respond differently when sharing is broken. The structural matrix Ω_{struct} depends only on the binning geometry and is blind to event occupancy entirely. When there is a genuine difference in phase-space acceptance between blocks, this can produce bin combinations absent from Ω_{struct} , breaking the hierarchy of Eq. (2.9) so that $\text{rank}(\Omega)$ exceeds $\text{rank}(\Omega_{\text{struct}})$; the effect is visible in Ω but not in Ω_{struct} . The empirical combination matrix Ω depends on which bin combinations are occupied, but not on how many events carry each, so it is blind to a violation caused only by missing events. In that case we can have $\text{rank}(\Omega) = \text{rank}(\Omega_{\text{struct}})$ even though the affected direction has acquired genuine variance. Only the spectrum of C_{stat} reflects that variance, lifting the corresponding eigenvalue out of the null space.

These two failure modes are illustrated with the toy model of Sec. 4. The first is demonstrated by extending the p_μ acceptance of the two-dimensional block beyond that of the p_μ marginal, so that events with $p_\mu > 1.2 \text{ GeV}/c$ populate the former but not the latter. Of the $O(5 \times 10^5)$ generated events only seven fall in this region, yet their presence is enough to raise $\text{rank}(\Omega)$ from 17 to 18, above $\text{rank}(\Omega_{\text{struct}}) = 17$, and to make the affected block total fall short of the others by exactly

those seven events. The rank signature is thus sensitive to even a single forbidden bin combination.

The second is demonstrated by removing a fraction of events from one block at random, so that the shared-event structure is broken without altering the binning. Figure 8 shows the eigenvalue spectrum of C_{stat} as this fraction is increased independently for blocks 1 (p_μ marginal) and 2 ($\cos \theta_\mu$ marginal). The formerly null eigenvalues, exactly zero under perfect sharing, are lifted into the bulk of the spectrum, filling the spectral gap that otherwise identifies the null space cleanly. The pattern set is unchanged, so $\text{rank}(\Omega) = \text{rank}(\Omega_{\text{struct}})$ throughout; only C_{stat} reflects the violation.

These complementary sensitivities provide a practical set of consistency checks rather than a reason to abandon the geometric construction. Before relying on the structural projection, one should verify that the spectral gap of C_{stat} is clean, that any difference between $\text{rank}(\Omega)$ and $\text{rank}(\Omega_{\text{struct}})$ is understood, and that the measured block totals are consistent within their correlated statistical uncertainties. A deficit, $\text{rank}(\Omega) < \text{rank}(\Omega_{\text{struct}})$, is expected when allowed bin combinations are simply unpopulated; this is the kinematic null case seen in Sec. 5, handled correctly by the empirical Ω . An excess, $\text{rank}(\Omega) > \text{rank}(\Omega_{\text{struct}})$, instead signals that events occupy structurally forbidden combinations, the hallmark of broken sharing. When the diagnostics are consistent with exact sharing, the a priori count of Sec. 2 applies as derived. When they are not, the structural null space over-counts the constrained directions, and projecting onto it would discard information; this signals that the binning or selection definitions should be reconciled across blocks, or that a treatment retaining the affected directions, such as marginalizing over nuisance parameters describing the differences (Sec. 3.3), is more appropriate.

7 Summary and conclusions

Multi-distribution cross-section measurements that share a common event sample contain exact linear constraints among their bin counts. When the statistical covariance accounts for the inter-distribution correlations, these constraints appear as a rank deficiency, and a global χ^2 that ignores them is ill-defined. The null directions either prevent inversion or, once lifted by systematic uncertainties, produce unphysically inflated values. We have presented the range-projected χ^2 , a goodness-of-fit statistic that restricts the test to the subspace carrying independent statistical information, yielding a χ^2 with $N_{\text{bins}} - N_{\text{null}}$ degrees of freedom.

This work makes two key contributions. First, we have shown that this rank deficiency is a predictable consequence of the shared-event structure rather than an incidental numerical degeneracy. The constrained directions are exactly those along which the data carry no statistical power, so removing them discards no physical information. Second, we have provided a prescription to compute the number of constrained directions a priori from the binning geometry, decomposing N_{null} into a structural contribution fixed by the bin edges and a kinematic contribution set by the phase-space occupancy. Because the rank is determined from the measurement geometry rather than from the systematics-contaminated covariance spectrum, the method requires no eigenvalue threshold on the full covariance, where no clean separation between null and physical directions needs to exist.

The method was validated in two settings. In an analytical toy model the projected statistic follows the expected $\chi^2(17)$ distribution, while the unprojected statistic is inflated to a mean of 30.6 and over-rejects the correct model in roughly one third of experiments; we traced the inflation

to a noncentrality on the directions lifted out of the null space by systematic uncertainties. In a simulated neutrino–argon measurement with unfolding and multi-source systematic uncertainties, the projected statistic gives 15.7/16 for the correct model and 154.4/16 for an incorrect one, recovering both calibration and discriminating power, whereas the unprojected statistic rejects both models indistinguishably. The two studies share five null directions of differing composition, structural in the toy and a mixture of structural and kinematic in the simulated measurement. We note that the projected statistic is mildly over-dispersed relative to the ideal χ^2 , with a false rejection rate of 5.7% at the nominal 5% level in the toy, which does not affect the conclusions.

The method assumes that the event-sharing structure is exact. Departures from this assumption are detectable through the spectral gap of the statistical covariance, the comparison of the empirical and structural ranks, and the consistency of the measured block totals, and we have given a corresponding set of diagnostics and remedies. Where the full forward model and nuisance parameterization are available, a binned likelihood fit handles the constraints automatically and is more robust to imperfect sharing; the range-projected χ^2 instead requires only the reported cross-block covariance, the binning, and the detector response, making it suited to the goodness-of-fit testing, model tuning, and generator comparison that reuse published measurements.

As multi-distribution measurements with shared events become more common, a correctly calibrated global goodness-of-fit test becomes a routine requirement for their interpretation. The algorithm for constructing the structural combination matrix is given in Appendix A, and a reference implementation is available at [?].

Acknowledgments

A Structural combination matrix algorithm

We describe the algorithm implemented to construct the structural combination matrix Ω_{struct} from the bin edge specifications of the multi-block measurement. Each row of Ω_{struct} is a unique indicator vector encoding one structurally allowed combination of bins, one per block, that a single event can simultaneously populate. The rank of this matrix determines the number of null directions, as described in Sec. 2.3.

The construction proceeds in three stages. First, the kinematic variables used across all blocks are classified as shared (appearing in two or more blocks) or private (appearing in exactly one block). This distinction is key to the efficiency of the algorithm. Shared variables create correlations between blocks, while private variables contribute bins that are independent of all other blocks.

Second, a common refinement is computed for each shared variable by taking the union of all bin edges from every block that uses it. The algorithm then enumerates the cells of this shared-variable refinement. For each cell, the midpoint in each shared variable is used to identify which bin (or set of bins) in each block is compatible with that cell. When the binning of a given block is fully determined by the shared variables (e.g., a 1D block whose variable is shared, or a 2D block whose slice and bin variables are both shared) the cell maps to a single bin. When a block uses a private variable, the shared cell constrains only part of the binning, and the block contributes multiple candidate bins corresponding to the different private-variable choices. A 1D block whose

variable is private contributes all of its bins; a 2D block with a shared slice variable and a private bin variable contributes all bins within the determined slice.

Third, the candidate bins from each block are combined. In the simplest case, the full Cartesian product of all candidate bins is expanded, and each element becomes a row of Ω_{struct} . However, when many blocks have private variables, this product can become very large. In that case, a representative subset of indicator vectors is used instead. A base combination is formed by selecting the first candidate bin from each block, and then the remaining candidates from each block are substituted one at a time while the others are held at their base values. This produces at most $1 + \sum_b (n_b - 1)$ rows per shared cell, where n_b is the number of candidate bins for block b . The representative set spans the same column space as the full Cartesian product, and therefore yields the same rank(Ω_{struct}) and null-space structure.

To understand why, note that any indicator vector from the full product can be written as the sum of indicator vectors from the representative set. Denoting the base combination as $\mathbf{e}_0$ and the single-block variations as $\mathbf{e}_j^{(k)}$ (block k switched to its j -th candidate), any full-product row that selects candidate j_k in each block satisfies

$$\mathbf{e}_{j_1, \dots, j_B} = \mathbf{e}_0 + \sum_k (\mathbf{e}_{j_k}^{(k)} - \mathbf{e}_0), \quad (\text{A.1})$$

since the blocks occupy disjoint columns. The full product therefore lies in the span of the representative rows, and the rank is preserved.

When the measurement involves mutually exclusive event selections, blocks belonging to different selections share no events and are processed independently. The algorithm is applied separately to each selection group, and the resulting matrices are vertically concatenated.

The cost of the algorithm is dominated by the enumeration of shared-variable cells, which scales as $\prod_{v \in \text{shared}} M_v$ where M_v is the number of refined bins for shared variable v . Since only shared variables enter this product, measurements with many blocks but few shared variables, such as a set of 1D distributions in different kinematic variables, are handled efficiently regardless of the total number of bins.

References

- [1] MICROBooNE collaboration, *First Simultaneous Measurement of Differential Muon-Neutrino Charged-Current Cross Sections on Argon for Final States with and Without Protons Using MicroBooNE Data*, *Phys. Rev. Lett.* **133** (2024) 041801 [[2402.19281](#)].
- [2] MICROBooNE collaboration, *Inclusive cross section measurements in final states with and without protons for charged-current ν_μ -Ar scattering in MicroBooNE*, *Phys. Rev. D* **110** (2024) 013006 [[2402.19216](#)].
- [3] MICROBooNE collaboration, *First Double-Differential Measurement of Kinematic Imbalance in Neutrino Interactions with the MicroBooNE Detector*, *Phys. Rev. Lett.* **131** (2023) 101802 [[2301.03706](#)].
- [4] MICROBooNE collaboration, *Multidifferential cross section measurements of $\nu\mu$ -argon quasielasticlike reactions with the MicroBooNE detector*, *Phys. Rev. D* **108** (2023) 053002 [[2301.03700](#)].

- [5] MicroBooNE collaboration, *Measurement of three-dimensional inclusive muon-neutrino charged-current cross sections on argon with the MicroBooNE detector*, *Phys. Lett. B* **870** (2025) 139939 [2307.06413].
- [6] T2K collaboration, *Characterization of nuclear effects in muon-neutrino scattering on hydrocarbon with a measurement of final-state kinematics and correlations in charged-current pionless interactions at T2K*, *Phys. Rev. D* **98** (2018) 032003 [1802.05078].
- [7] T2K collaboration, *First T2K measurement of transverse kinematic imbalance in the muon-neutrino charged-current single- π^+ production channel containing at least one proton*, *Phys. Rev. D* **103** (2021) 112009 [2102.03346].
- [8] NOvA collaboration, *Measurement of the double-differential muon-neutrino charged-current inclusive cross section in the NOvA near detector*, *Phys. Rev. D* **107** (2023) 052011 [2109.12220].
- [9] NOvA collaboration, *Measurement of $d^2\sigma/d|\vec{q}|dE_{avail}$ in charged current ν_μ -nucleus interactions at $\langle E_\nu \rangle = 1.86$ GeV using the NOvA Near Detector*, *Phys. Rev. D* **111** (2025) 052009 [2410.05526].
- [10] NOvA collaboration, “Measurement of the double-differential cross section of muon-neutrino charged-current interactions with low hadronic energy in the NOvA Near Detector.” 10, 2024.
- [11] T2K collaboration, *Simultaneous measurement of the muon neutrino charged-current cross section on oxygen and carbon without pions in the final state at T2K*, *Phys. Rev. D* **101** (2020) 112004 [2004.05434].
- [12] MINERvA collaboration, *Simultaneous Measurement of ν_μ Quasielasticlike Cross Sections on CH, C, H₂O, Fe, and Pb as a Function of Muon Kinematics at MINERvA*, *Phys. Rev. Lett.* **130** (2023) 161801 [2301.02272].
- [13] MINERvA collaboration, *Simultaneous Measurement of Muon Neutrino ν_μ Charged-Current Single π^+ Production in CH, C, H₂O, Fe, and Pb Targets in MINERvA*, *Phys. Rev. Lett.* **131** (2023) 011801 [2209.07852].
- [14] T2K collaboration, *First combined measurement of the muon neutrino and antineutrino charged-current cross section without pions in the final state at T2K*, *Phys. Rev. D* **101** (2020) 112001 [2002.09323].
- [15] S. Gardiner, “Mathematical methods for neutrino cross-section extraction.” 2024.
- [16] G. D’Agostini, *A Multidimensional unfolding method based on Bayes’ theorem*, *Nucl. Instrum. Meth. A* **362** (1995) 487.
- [17] W. Tang, X. Li, X. Qian, H. Wei and C. Zhang, *Data Unfolding with Wiener-SVD Method*, *JINST* **12** (2017) P10002 [1705.03568].
- [18] C. Wilkinson et al., *Testing charged current quasi-elastic and multinucleon interaction models in the NEUT neutrino interaction generator with published datasets from the MiniBooNE and MINERvA experiments*, *Phys. Rev. D* **93** (2016) 072010 [1601.05592].
- [19] MINERvA collaboration, *Tuning the GENIE Pion Production Model with MINERvA Data*, *Phys. Rev. D* **100** (2019) 072005 [1903.01558].
- [20] GENIE collaboration, *Neutrino-nucleus CC0 π cross-section tuning in GENIE v3*, *Phys. Rev. D* **106** (2022) 112001 [2206.11050].
- [21] MicroBooNE collaboration, *Measurement of single- and double-differential cross sections for mesonless charged-current muon neutrino interactions on argon with final-state protons using the MicroBooNE detector*, *Phys. Rev. D* **112** (2025) 112004 [2403.19574].

- 837 [22] C. Rao and S. Mitra, *Generalized Inverse of Matrices and Its Applications*, Probability and Statistics
838 Series, Wiley (1971).
- 839 [23] T2K collaboration, *First measurement of muon neutrino charged-current interactions on hydrocarbon
840 without pions in the final state using multiple detectors with correlated energy spectra at T2K*, *Phys.
841 Rev. D* **108** (2023) 112009 [[2303.14228](#)].
- 842 [24] P. Stowell et al., *NUISANCE: a neutrino cross-section generator tuning and comparison framework*,
843 *JINST* **12** (2017) P01016 [[1612.07393](#)].
- 844 [25] C. Andreopoulos et al., *The GENIE Neutrino Monte Carlo Generator*, *Nucl. Instrum. Meth. A* **614**
845 (2010) 87 [[0905.2517](#)].
- 846 [26] T. Golan, J.T. Sobczyk and J. Zmuda, *NuWro: the Wroclaw Monte Carlo Generator of Neutrino
847 Interactions*, *Nucl. Phys. B Proc. Suppl.* **229-232** (2012) 499.
- 848 [27] A.S. Meyer, M. Betancourt, R. Gran and R.J. Hill, *Deuterium target data for precision
849 neutrino-nucleus cross sections*, *Phys. Rev. D* **93** (2016) 113015 [[1603.03048](#)].
- 850 [28] J. Nieves, I. Ruiz Simo and M.J. Vicente Vacas, *Inclusive Charged-Current Neutrino-Nucleus
851 Reactions*, *Phys. Rev. C* **83** (2011) 045501 [[1102.2777](#)].
- 852 [29] D. Rein and L.M. Sehgal, *Neutrino Excitation of Baryon Resonances and Single Pion Production*,
853 *Annals Phys.* **133** (1981) 79.
- 854 [30] S. Dytman, Y. Hayato, R. Raboanary, J.T. Sobczyk, J. Tena Vidal and N. Vololoniaina, *Comparison of
855 validation methods of simulations for final state interactions in hadron production experiments*, *Phys.
856 Rev. D* **104** (2021) 053006 [[2103.07535](#)].